## Last time

Introduced graph div/grad

Today → Flows on graphs

- higher ~~order~~ operators
- graph Laplacians

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## Formalizing

<u>K-chain</u> $C^K = [v_1, \dots v_K]_{v_i \in V}$ anti-symmetric wrt boundary swaps

i.e.

$$[v_1, v_2] \quad - \quad [v_2, v_1]$$

<u>boundary</u> $\partial_K : C^{K+1} \to C^K$

$$\partial_K [v_1, \dots v_{K+1}] = \sum_{i=1}^{K+1} (-1)^{i-1} [n_1, \dots \hat{n}_i, \dots n_{K+1}]$$

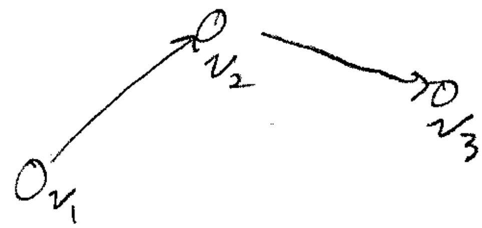
leave out

<u>ex</u>

$$\partial [v_1, v_2] = (-1)^0 [v_2] + (-1)^1 [v_1] = v_2 - v_1$$

We can use this to turn geometry into algebra

$$S = [v_1, v_2] + [v_2, v_3]$$




$$\partial S = v_2 - v_1 + v_3 - v_2$$

$$= v_3 - v_1$$

<u>Thm</u> Exact sequence property

$$\partial_{k-1} \circ \partial_k = 0$$

<u>Pf</u> Let $c \in C^{k+1}$

$$\partial_{k-1} \partial_k c = \sum_{i=1} (-1)^{i-1} \partial_{k-1} [n_1, \ldots \hat{n}_i, \ldots n_k]$$

$$= \sum_{i,j} (-1)^{i-1} (-1)^{j-1} [n_1, \ldots \hat{n}_i \ldots \hat{n}_j \ldots n_{k+1}]$$

antisymmetric under i,j swap

$$= \sum_{i,j} \underbrace{(-1)^{i+j-2}}_{\text{sym}} \underbrace{\alpha_{ij}}_{\text{antisym}}$$

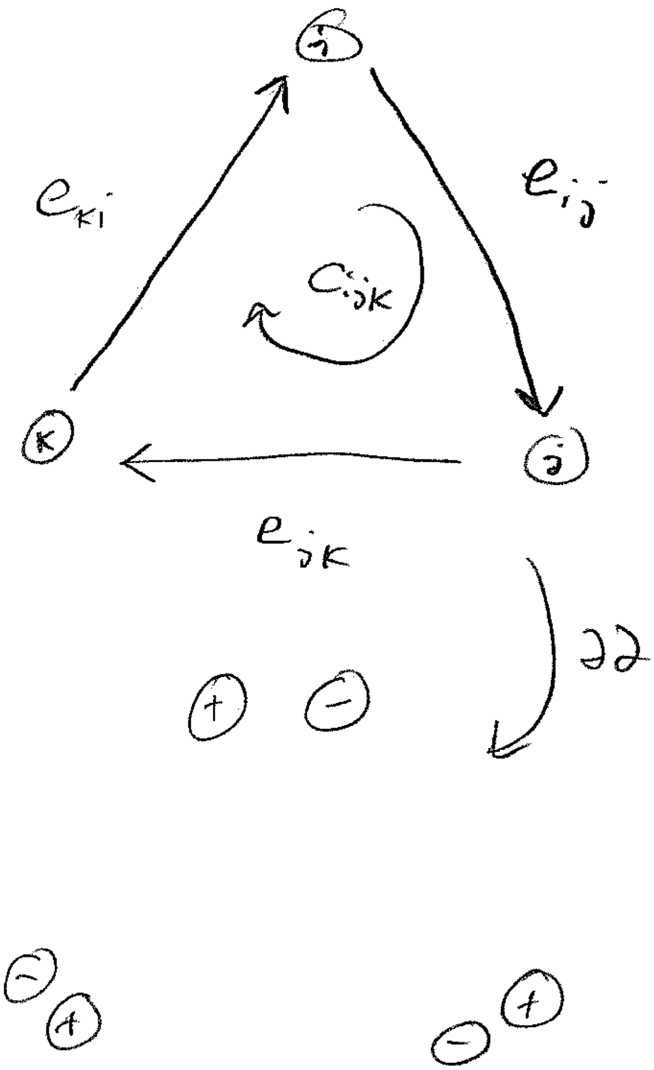
$$= 0$$

# Geometry

$$c_{ijk} = [v_i, v_j, v_k]$$

$$\partial c_{ijk} = e_{ij} + e_{jk} + e_{ki}$$
$$= [v_i, v_j] + [v_j, v_k] + [v_k, v_i]$$

$$\partial\partial c_{ijk} = v_j - v_i + v_k - v_j + v_i - v_k = 0$$




To define coboundary (graph div/grad/curl)
swap chains + cochains

For $f \in C_k$ $\qquad \delta_k : C_k \to C_{k+1}$

$$\delta_k f = \sum_i (-1)^i f(v_1, \ldots \hat{v}_i, \ldots v_k)$$

ex grad $\qquad \delta_0 \phi_{ij} = \phi_j - \phi_i$

curl $\qquad \delta_1 A_{ijk} = A_{ij} + A_{jk} + A_{ki}$

curl $\circ$ grad $= 0$

$$\delta_1 \delta_0 \phi = \delta_0 \phi_{ij} + \delta_0 \phi_{jk} + \delta_0 \phi_{ki}$$

$$= \phi_j - \phi_i + \phi_k - \phi_j + \phi_i - \phi_k$$

$$= 0$$

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<u>Codifferentials</u> $\qquad$ Define $(\delta^* f, g) = (f, \delta g)$ in $l_2$ inner product

<u>Recall</u> $\qquad \delta_0 = \text{grad} \qquad \delta_0^* = \text{div}$

$\qquad\qquad \delta_1 = \text{curl} \qquad \delta_1^* = \text{curl}$

$$C_0 \underset{\delta_0^*}{\overset{\delta_0}{\rightleftarrows}} C_1 \underset{\delta_1^*}{\overset{\delta_1}{\rightleftarrows}} C_2$$

Thm $\quad \delta^* \circ \delta^* = 0$

pf $\quad \langle f, \delta_k^* \delta_{k+1}^* g \rangle \quad$ for $f \in C^{k-2}$, $g \in C^k$

$= \langle \delta_k f, \delta_{k+1}^* g \rangle$

$= \langle \underline{\delta_{k+1} \delta_k f}, g \rangle$

$= 0$

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We now have a complete graph analog to div/grad/curl. Same as vector calc we can use them to solve Laplacian problems

$$\Delta_k = \delta_{k-1} \delta_{k-1}^* + \delta_k^* \delta_k$$

$$\begin{tikzcd} C^{k-1} \arrow[r, shift left, "\delta_{k-1}"] & C^k \arrow[l, shift left, "\delta_{k-1}^*"] \arrow[r, shift left, "\delta_k"] & C^{k+1} \arrow[l, shift left, "\delta_k^*"] \end{tikzcd}$$

ex $\quad k=0 \qquad \delta_0^* \delta \longrightarrow \nabla \cdot \nabla$

$\quad k=1 \qquad \delta_1^* \delta_1 + \delta_0 \delta_0^* \longrightarrow -\nabla \times \nabla \times + \nabla \nabla \cdot$

## ex application

In vector calc

$$F = \nabla \times A + \nabla \phi$$

How to find? Hodge Projection

$$\nabla \cdot F = \cancel{\nabla \cdot \nabla \times A} + \nabla^2 \phi$$

$$\nabla \times F = \nabla \times \nabla \times A + \cancel{\nabla \times \nabla \phi}$$

→ 2 PDEs to solve

On a graph:

Let $f \in C_K$

$$f = \delta_{K-1} \phi + \delta_K^* A$$

$$\delta_K f = \cancel{\delta_K \delta_{K-1}} \phi + \delta_K \delta_K^* A$$

$$\delta_{K-1}^* f = \delta_{K-1}^* \delta_{K-1} \phi + \cancel{\delta_{K-1}^* \delta_K^*} A$$

Pause → Why useful?

- Graph encoders (Bronstein GRAND)
- DAG discovery (NO TEARS)
- Recommender systems
- Circuit models

We have the machinery for analysis

① Turn into variational problem

$L^w[u]_i := \sum_j W_{ij}(u_j - u_i) = f_i$ weighted graph Laplacian

$f, u \in C_0$, $W_{ij} = W_{ji}$

Let $v \in C_0$

Test → $\langle v, L^w[u] \rangle = \langle v, f \rangle$

$$= \sum_{i,j} W_{ij}(u_j - u_i) v_i$$

$$= \frac{1}{2} \sum_{i,j} W_{ij}(u_j - u_i) v_i + \frac{1}{2} \sum_{i,j} W_{ij}(u_i - u_j) v_j$$

$$= \frac{1}{2} \sum_{i,j} W_{ij}(u_j - u_i)(v_j - v_i)$$

weighted inner product on $C_1$

$$:= a(u,v) = \frac{1}{2}(\delta v)^T W (\delta v)$$

Lax-Milgram isn't just for fields → applies equally here!

① $\|u\|_E^2 = a(u,u) \overset{?}{\geq} \alpha \langle u,u \rangle = \alpha \|u\|^2$

② $|a(u,v)| \leq \gamma \|u\| \|v\|$

To get $\alpha$ and $\gamma$, we need to understand eigenvalues

# Rayleigh Quotients + Eigenanalysis

Let $M = M^T$ w/ orthogonal eigenpairs $(\lambda_i, v_i)$

$v_i^T v_i = \delta_{ij}$

So $M v_i = \lambda v_i$ (def of eig)

$v_i^T M v_i = \lambda v_i^T v_i$

$= \lambda$

<u>Def</u> Rayleigh Quotient

$$R(M,x) = \frac{x^T M x}{x^T x}$$

$\leftarrow$ energy norm

$\leftarrow$ norm we want control over

Expanding $x$ in eigenbasis

$$x = \sum_i \hat{y}_i v_i$$

where $\hat{y}_i = \langle v_i, x \rangle$

Then $R(M,x) = \dfrac{\sum_i \lambda_i \hat{y}_i^2}{\sum_j \hat{y}_j^2}$

Min/Max of $R$ gives min/max eigs

$$\min_x R(M,x) = \lambda_{min} = \lambda_{min} \frac{\sum \hat{y}_i^2}{\sum \hat{y}_j^2} \leq \frac{\sum \lambda_i \hat{y}_i^2}{\sum \hat{y}_i^2} \leq \lambda_{max} = \max_x R(M,x)$$

Lets estimate min eig for graph Laplacian | Spectral Graph Theory Fan Chung

$$\lambda_1 = \min_{f \perp 1} \frac{\sum (f_i - f_j)^2}{\sum f_k^2}$$

(1) ~~the constant vector has zero~~ eigenvalue

$$\cancel{v_0 = 1 \qquad \lambda_0 = 0}$$

By orthogonality $\langle v_0, f_{min} \rangle = 0 \rightarrow \sum_j f_{min} = 0$

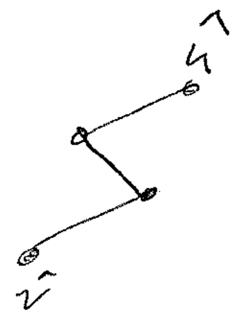
(2) Pick $\hat{v}$ s.t. $|f(\hat{v})| = \max_v |f(v)|$

Because $\sum f_i = 0$, we can find a $\hat{u}$ such that $f(\hat{u}) f(\hat{v}) < 0$

(3) Define

$$D = \max_{i,j \in V} d(v_i, v_j)$$ diameter of graph

Let $P$ be a path connecting $\hat{u}$ to $\hat{v}$ with $m \leq D$ edges




$$f(\hat{v}) - f(\hat{u}) = \sum_k f(v_{k+1}) - f(v_k)$$

(4) C-S $$(f(\hat{v}) - f(\hat{u}))^2 \leq m \sum_P (f(v_{k+1}) - f(v_k))^2$$

$$\leq D \sum_{i,j} (f_j - f_i)^2$$

Numerator of RQ

Because $f(\hat{u})$ & $f(\hat{v})$ have opposite signs

and $|f(\hat{v})| = \max |f(v)|$

$$\left( f(\hat{v}) - f(\hat{u}) \right)^2 \geq f(\hat{v})^2$$

$$\Rightarrow \quad f(\hat{v})^2 \leq D \sum_{i,j} (f_j - f_i)^2$$

(5) For denominator, since $\hat{v}$ is max

$$\sum_{k=1}^{N} f_k^2 \leq N f(\hat{v})^2$$

(4) + (5)

$$\lambda_1 = \frac{\sum (f_i - f_j)^2}{\sum f_i^2} \geq \frac{f(\hat{v})^2 / D}{N f(\hat{v})^2} = \frac{1}{ND}$$

So $\alpha$ in Lax Milgram is $\frac{1}{ND}$.